

07 - Managed Databases

CSCI 112 - Contemporary Databases

Maiko Francisco V Rugeria, MTM



Learning Objectives

- Understand what managed databases are and how they are different from traditional databases
- Explain what DynamoDB and MongoDB Atlas can do and identify use cases where it is applicable
- Learn how to identify the proper partition and sort keys when modeling data in DynamoDB

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Scenario for the Day



Context Setting

- Jigu Telecom, The Coco Group's telecommunications company partner in South Korea, would like to migrate their website detection logs to MongoDB.
- Currently, an RDBMS type database holds their data on subscriber website access. It's beginning to create bottlenecks for their analytics related micro services, so they thought it was time to migrate to a NoSQL based database.

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Traditional vs Managed Databases



Traditional Database Deployment Method (On Prem)

- Choose a database
- Plan for your database demand
- Setup on premise servers and install necessary operating systems and software
- Setup fault tolerance and high availability
- Use database
- Update servers and database to make sure operations are seamless

App optimization
Scaling
High availability
Database backups
DB s/w patches
DB s/w installs
OS patches
OS installation
HW Maintenance
Rack and stack
Power, HVAC, net

You



Traditional Database Deployment Method (Cloud)

- Choose a database
- Plan for your database demand
- Spawn server and install necessary software
- Setup fault tolerance and high availability
- Use database
- Update servers and database to make sure operations are seamless

App optimization

Scaling

High availability

Database backups

DB s/w patches

DB s/w installs

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You

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**Cloud
Provider**



Managed Database Deployment Method

- Choose a database
- Use database

App optimization

} **You**

Scaling

High availability

Database backups

DB s/w patches

DB s/w installs

**Cloud
Provider**

OS patches

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MongoDB Atlas



MongoDB Atlas

- MongoDB Atlas is a managed Cloud Deployment created by MongoDB themselves.
- It allows users to spawn database on demand with specific sizes and features, depending on what they need.
- You only pay for what you use. (Spawning a database instance is considered usage unless turned off)
- Go to mongodb.com to try it out!



MongoDB Atlas Modeling and Usage

- Data Modeling in MongoDB Atlas is the same as MongoDB.
- Queries and other CRUD functions are also the same.
- The difference is how you connect to it. You use a connection string that looks like this:
`mongodb+srv://<username>:<password>@csci112jigutelecom.kipbi.mongodb.net/`



Connecting Pymongo to Atlas

```
def openConnection():  
    url = 'mongodb+srv://<username>:<password>@csci112jigutelecom.kipbi.mongodb.net/'  
    conn = MongoClient(url)  
    return conn
```

Instead of the host IP, you only need to provide the Atlas URL. Take note that you need a username and password to be able to connect.



Benefits of MongoDB Atlas

- MongoDB Atlas is ACID compliant, even if it is a NoSQL document.
- Security concern is only up to the application level. MongoDB ensures that Atlas databases are secured by default.
- You can run serverless functions in MongoDB Atlas.
- Multi-region and multi-cluster setup can be done in a few clicks.
- Scaling up and out can be done in a few clicks.

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DynamoDB



DynamoDB

- DynamoDB is the managed NoSQL database created by AWS.
- It allows usage of a NoSQL database without any setup.
- It provides extremely high performance, more than 10 trillion requests per day with peaks greater than 20 million requests per second.
- It scales horizontally automatically, but this process is abstracted from the users.
- DynamoDB is serverless. You only pay for what you use and requires no cold starts, no patching, and no maintenance downtimes.

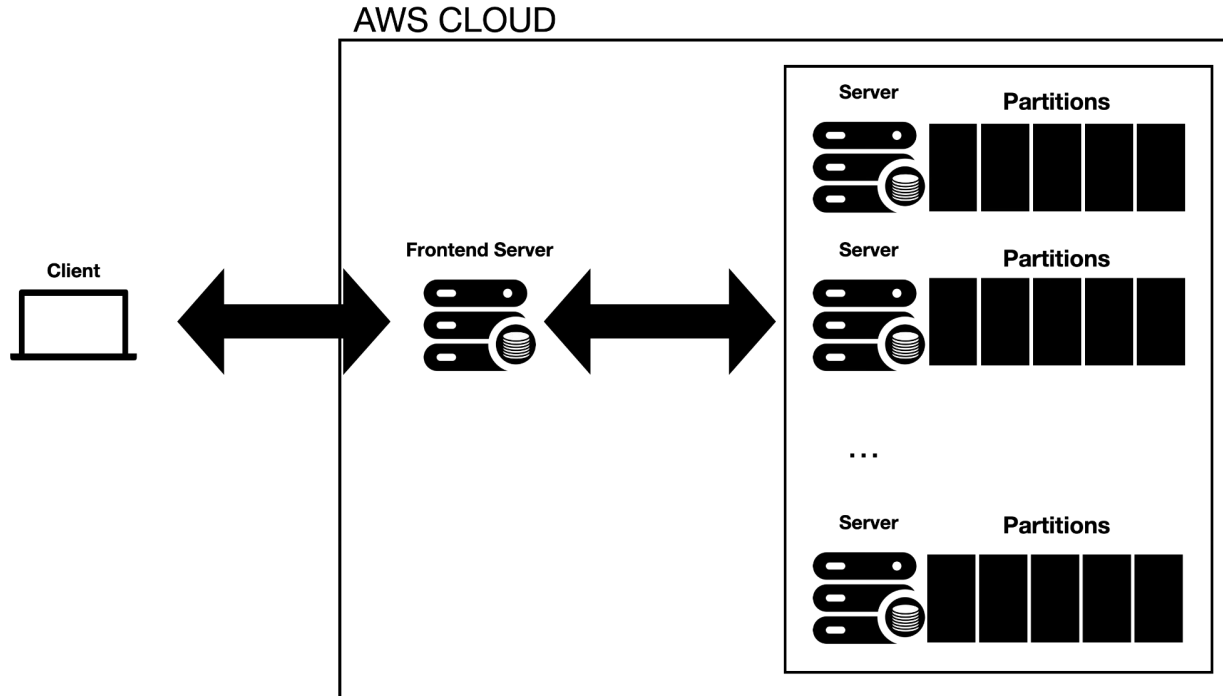


DynamoDB Replication

- There are 3 options to replicate data in DynamoDB:
 - ▷ On-demand backup and restore – manual backup in DynamoDB, backup data when you need to (e.g. before major deployments)
 - ▷ Point-in-time recovery – Instead of just snapshots, the DynamoDB keeps a transaction log/journal of changes that can be used to restore data from 7-35 days.
 - ▷ DynamoDB Global Tables – Automatic replication data across AWS regions.



Architecture





DynamoDB Partitions

- Partitions are locations in your DynamoDB servers that are not presented in any client layer.
- DynamoDB manages the partitions and the data that come into DynamoDB



DynamoDB Modeling: Components

- Primary Key - a primary key identifies records. It is composed of a partition key and a sort key
- Partition Key - used to partition records. This should be high in cardinality.
- Sort Key - used to sort records inside a partition. This should also be high in cardinality.
 - ▷ Combining the partition key and sort key will result to your primary key.
 - ▷ The sort key is optional.
 - ▷ Cannot be changed.

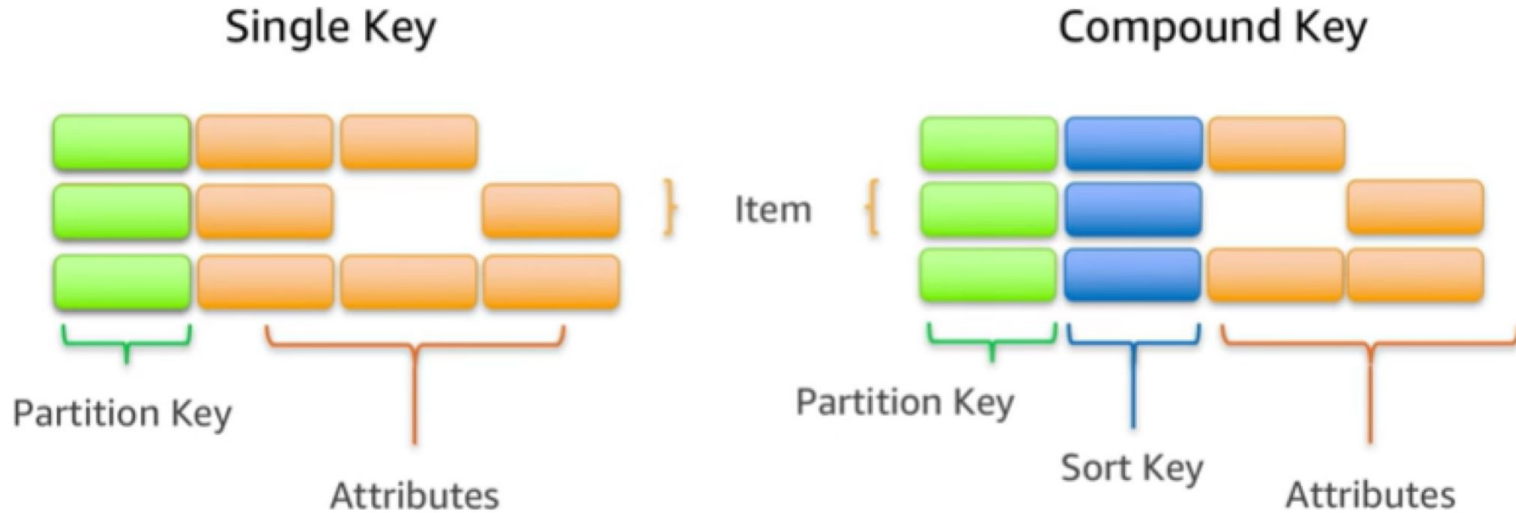


DynamoDB Modeling: Choosing a Partition Key

- Must be high cardinality
- Must be known to the application (access patterns)



DynamoDB Modeling: Single vs Compound Key





DynamoDB Query Behaviors: Query vs Scan

As data grows, table partitioned by key

QUERY by Key to find items efficiently
SCAN to find items by any attribute



DynamoDB Data Modeling Procedure

1. Identify your entities and relationships
2. Identify your access patterns
3. Choose between single and multiple table design
4. Choose a partition key and a sort/range key
5. Lay down your fields according to your chosen table design in #3



DynamoDB Secondary Index

- In case your access patterns change and you do not want a full scan of the table, you can create a secondary index on the fields that you will use for the new access patterns.
- There are 2 types of secondary indexes:
 - ▷ Global - Created on the base table
 - ▷ Local - Created based on a partition key



DynamoDB Secondary Index Pitfalls

- Creating a secondary index creates a replica of the base table that uses the new field as partition key (additional cost).
- Any change on the original table will eventually be reflected in the replica table.
- This doubles all writing costs.
- Race conditions may happen because of eventual consistency, wherein the secondary index returns stale data.

End.

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